

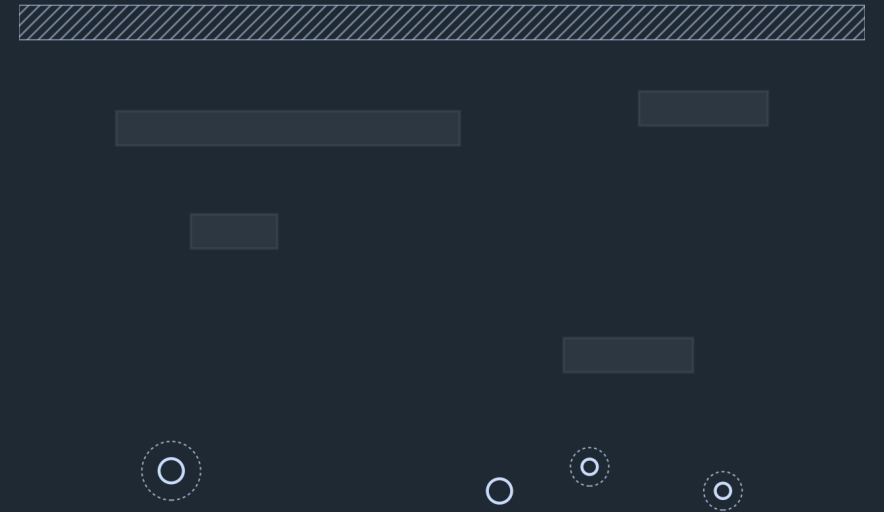
CrossMEP: A Tiered Synthetic Dataset of Multi-Trade MEP Cross-Sections

A support-design problem distribution for learning-based structural support assembly synthesis

Lavinia Pedrollo¹ • David Gvadzabia² • Torben Graeber³ • Martin Fischer¹

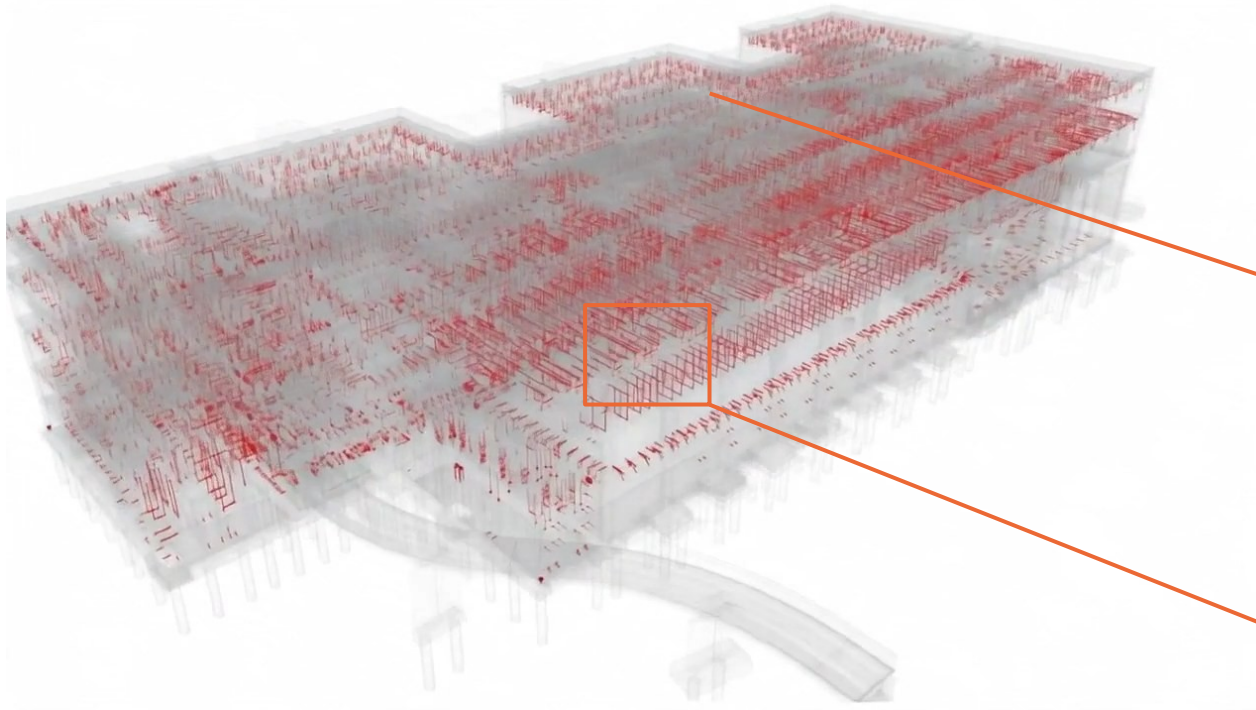
¹ Stanford University, CIFE ² Lafayette College ³ Hilti AG

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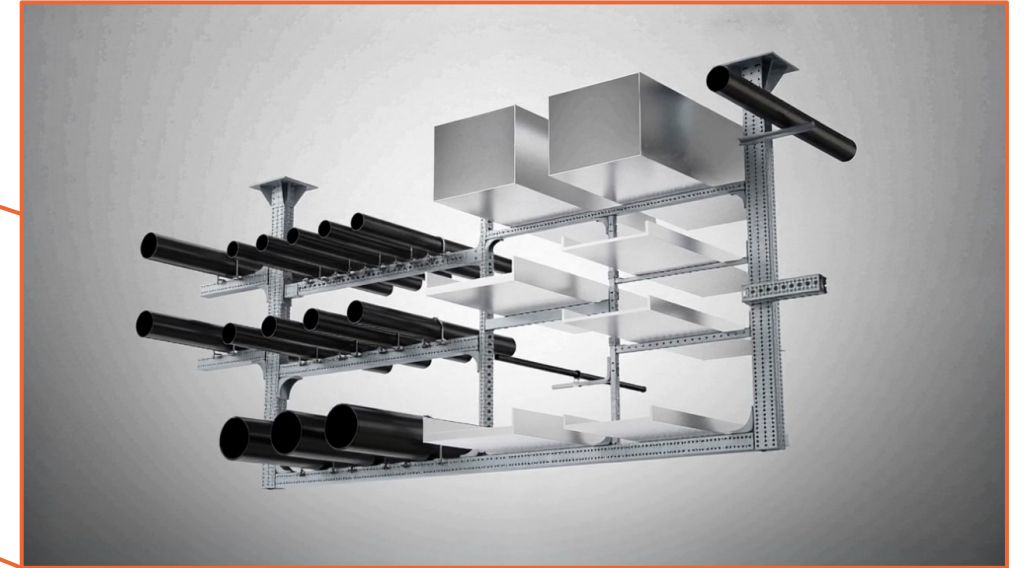


a generated C8 context: eight services on three rows

A hospital needs about 10,000 support assemblies, each designed by hand



1 unit = 1 structural support assembly (SSA)



$\approx 10,000$

support assemblies in one 200,000 sq ft hospital

20 min – 2 h

of engineering for each assembly, designed one by one

$\approx \frac{1}{4}$

of MEP design effort; on the order of \$600K per project

Approximate practitioner estimates.

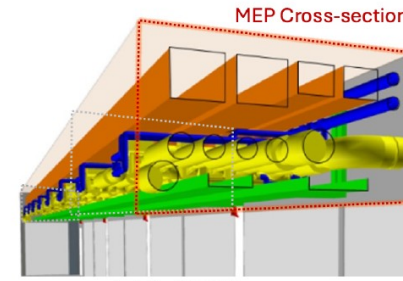
Synthesizing an assembly from a catalog and a section has resisted automation

The synthesis problem

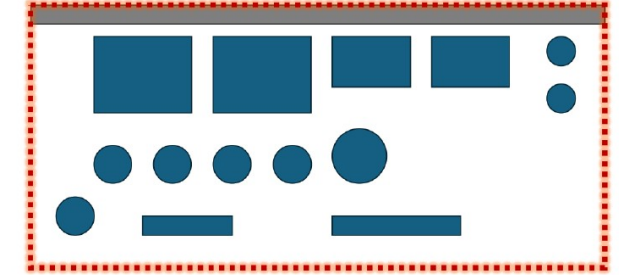
Given a fixed catalog of components and the MEP context: the services crossing the hanger and the surface they hang from.

Find a feasible assembly that carries the services to the structure.

Tools coordinate the model and check a design; choosing the topology and the parts is still done by hand.



(a) Full MEP route



(b) One CrossMEP context

Design starts after coordination: at each hanger the designer works from the section across the run, not from the whole model.

1. Geometric judgment

Engineers choose an assembly's topology from experience.

2. Rule explosion

Code, load and material constraints interact across thousands of combinations.

3. Catalog volatility

Catalogs change faster than rule-based systems can be rewritten.

Learning to design supports needs many problems, and none were public

Why it has to be synthetic



Scarcity

Project models are proprietary; the few open ones are not organised around supports and often export no element sizes.



Coverage

One project is one narrow slice: one building type, one trade mix. A method tuned to it measures fit to that project.



Control

A seeded generator gives unlimited problems with a known distribution, difficulty by construction and held-out seeds.

CrossMEP at a glance

7,000

contexts in four splits
on disjoint seeds

31,484

elements: pipes, cable
trays, ducts and
conduits; six trades

C1–C8

eight difficulty tiers by
exact element count

0 labels

the brief, not the
answer: no support-
assembly information

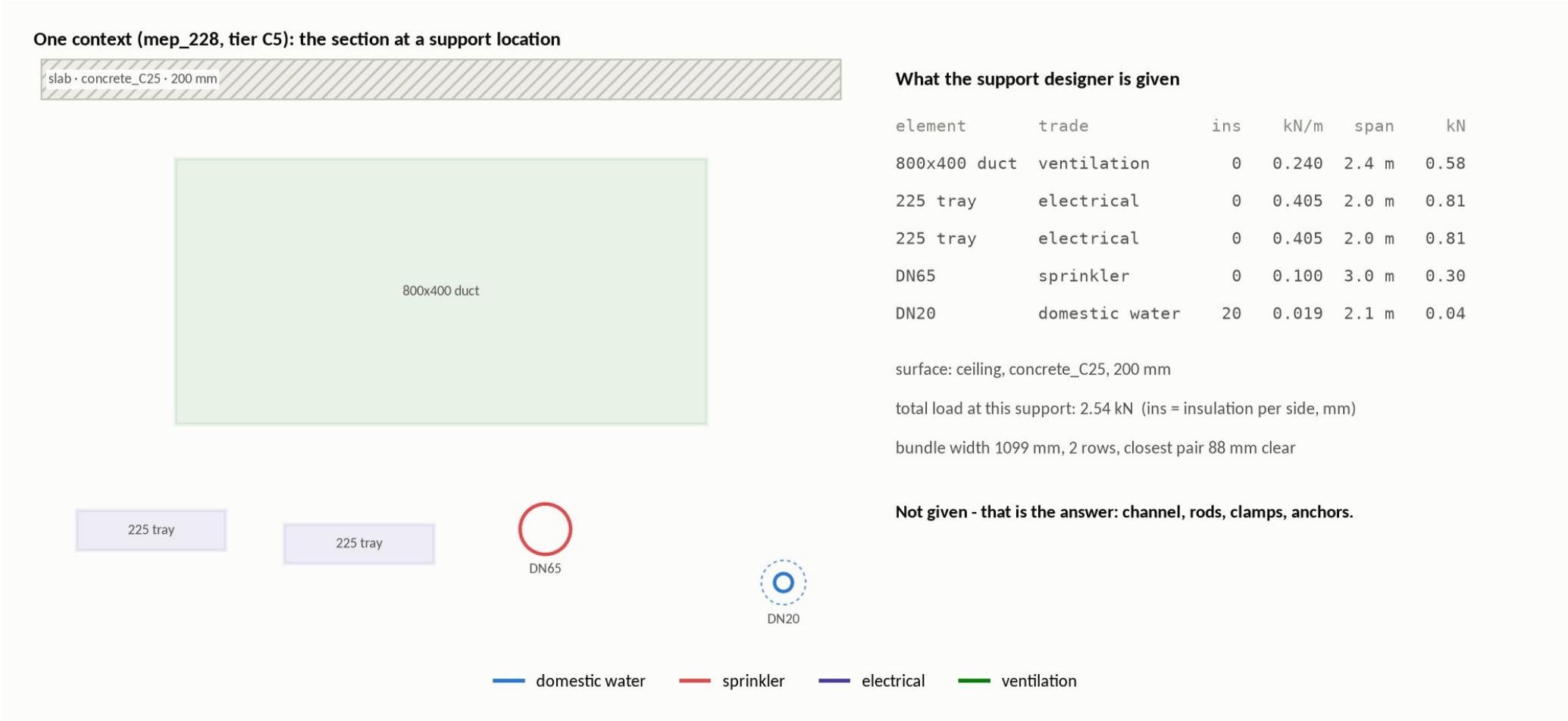
Traced

every constant with its
source, or declared a
design choice

Open

data CC BY 4.0, code
MIT; a seeded,
deterministic generator

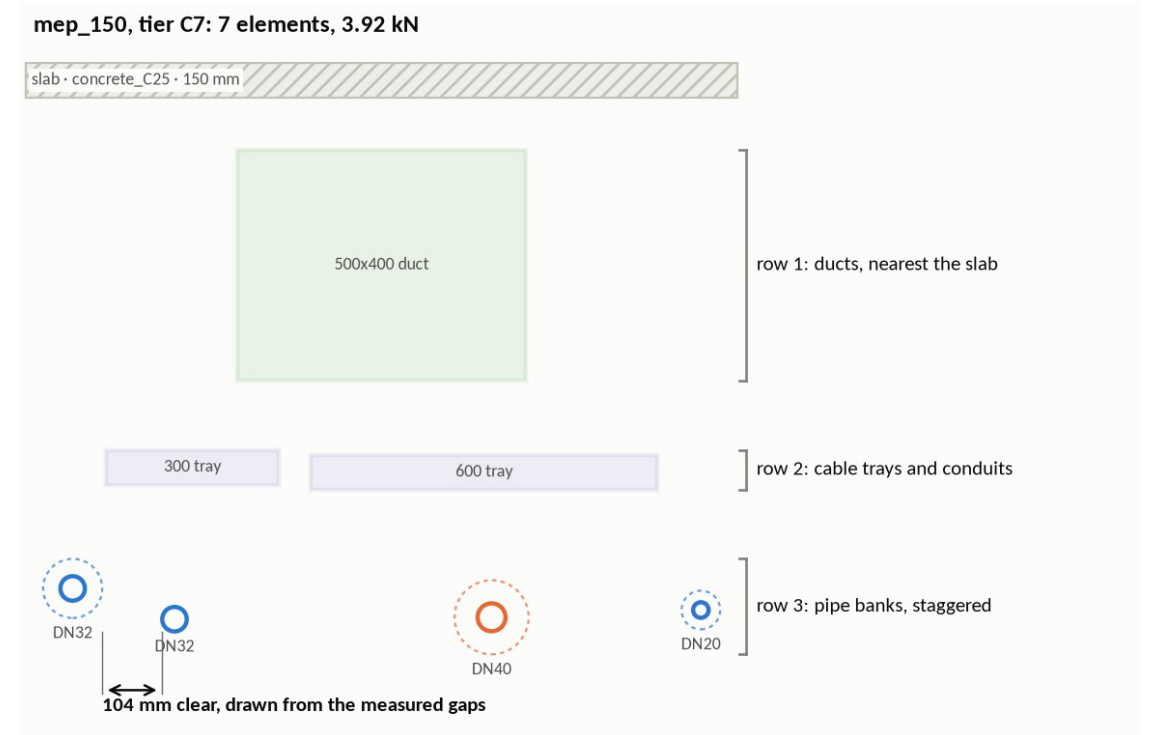
We generate what a designer receives: the section at one hanger



Per element: kind, service and trade · bare size · insulation per side · load per metre × the span it was sized at = load at the support · position along and out from the surface. Per context: slab or wall, substrate, thickness. Absent by design: channel, rods, clamps, anchors — and any “correct” answer, since a feasible support depends on the catalog you build from.

Sections follow trade practice and measured spacing, not random shapes

- 1 Tier.** Cn fixes the exact element count, $n = 1 \dots 8$.
- 2 Surface.** Ceiling or wall (78 / 22 %); concrete slab or wall, 150–300 mm thick.
- 3 Services.** Grouped as trades run: hot + cold pairs, flow + return, chilled banks, sprinkler mains, 1–2 trays, ducts, conduit groups of 2–6.
- 4 Rows.** Bulky first: ducts nearest the surface, then trays and conduits, then pipes.
- 5 Spacing.** Gaps drawn from gaps measured on a built project (at least 25 mm); stagger within rows; electrical above wet on walls.
- 6 Check.** Every pair at least 25 mm clear; rows centred; sizes, insulation, loads and positions recorded.



One generated benchmark context, three rows by priority; the gap marked is one draw from the measured distribution. Every context is seeded: the same seed gives the same file, byte for byte.

Sizes, spans and loads come from standards; the choices are declared

Element	Sizes	Load basis
Pipes DN15–100	EN 10220 / EN 10255 medium series	steel + water × ASME B31.1 water-service span (published points only)
Insulation	GEG Anlage 8 (heated); 30/50 mm condensation control (chilled)	—
Cable trays 150–600	IEC 61537 systems	full tray: 50 kg/m at 300 mm datum, × 2.0 m
Ducts	EN 1505 rectangular; EN 1506 round	manufacturer duct-weight table incl. flanges, × 2.4 m
Conduits Ø20–50	IEC 61386-1, parallel groups of 2–6	steel tube + 40 %-of-bore cable fill, × 2.0 m

Layout conventions

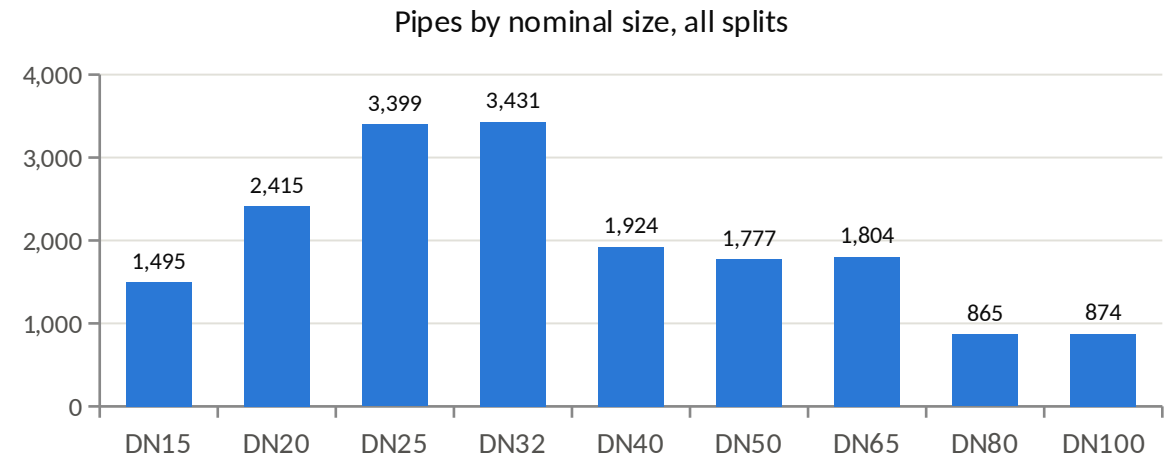
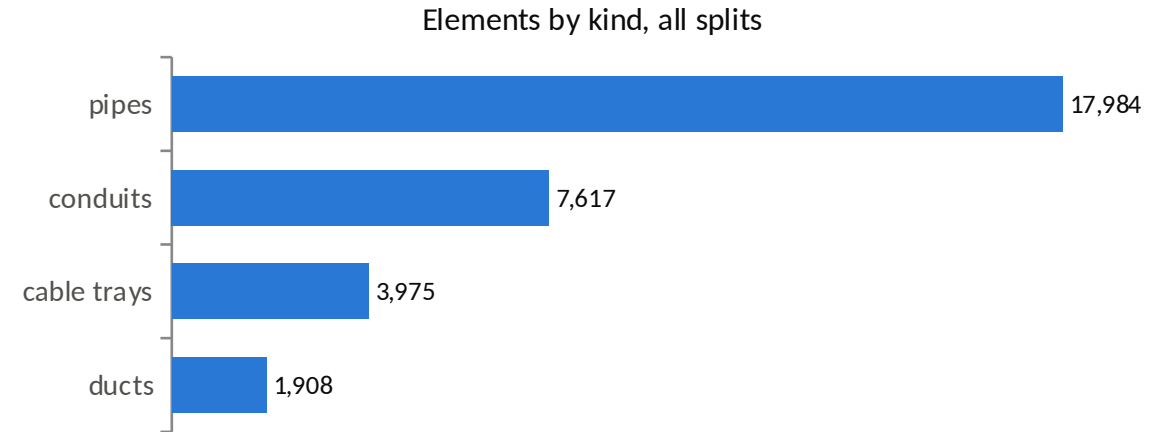
- Bulky services nearest the slab: ducts, then containment, then pipes
- Trades run in banks: hot + cold pairs, flow + return, conduit groups
- Electrical kept above wet services on walls (drip)
- Clear gaps drawn from a distribution measured on a built project; 25 mm minimum
- Within-row stagger calibrated to measured elevation spread

Every constant carries a source and a status (verified, practice-cited, or a declared design choice) in the repository. Loads are derived in code from those sources, so each number can be traced and checked.

7,000 contexts in four splits; the test seeds are never trained on

split	contexts	elements	seed
train	5,000	22,500	1000
validation	500	2,242	2000
test	500	2,242	3000
benchmark	1,000	4,500	42
total	7,000	31,484	

- Tiers assigned round-robin: 625 per tier in train, 125 in the benchmark
- Four in five contexts hang from a ceiling, one in five from a wall
- Six trades: electrical 11,592, domestic water 5,889, heating 5,766, chilled water 4,129, sprinkler 2,200, ventilation 1,908 elements
- Plain JSON, with a JSON Schema, Croissant metadata and a datasheet

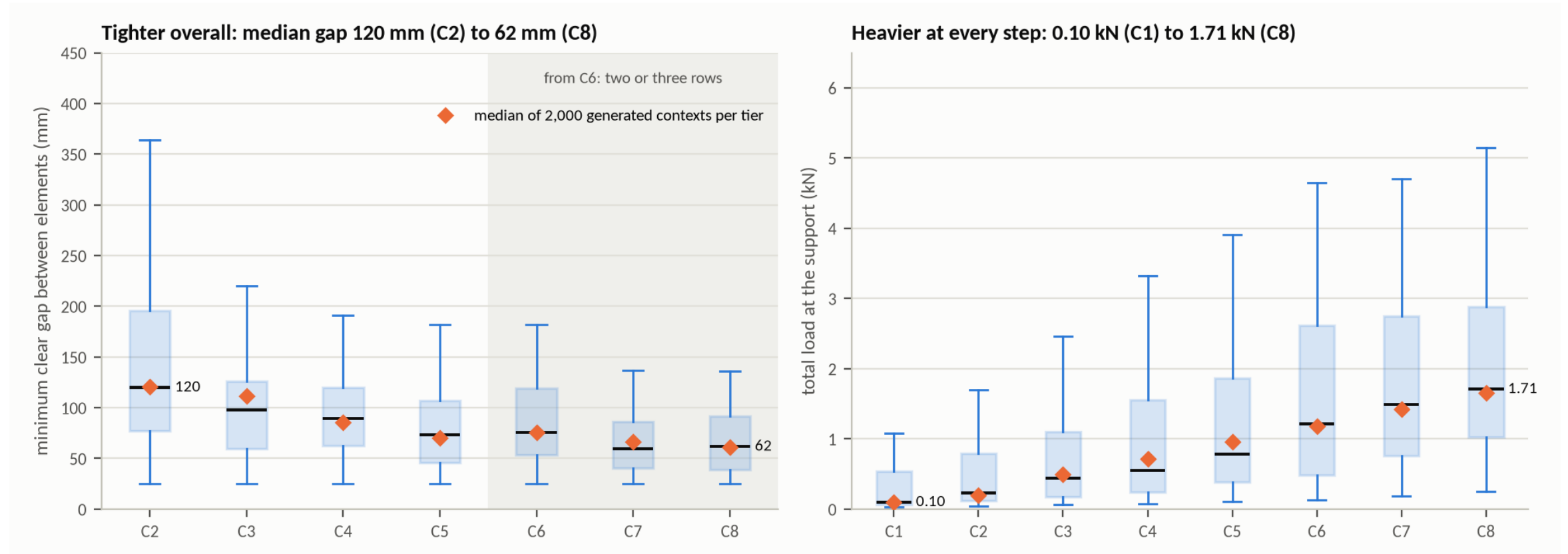


Difficulty is one number: tier Cn holds exactly n elements



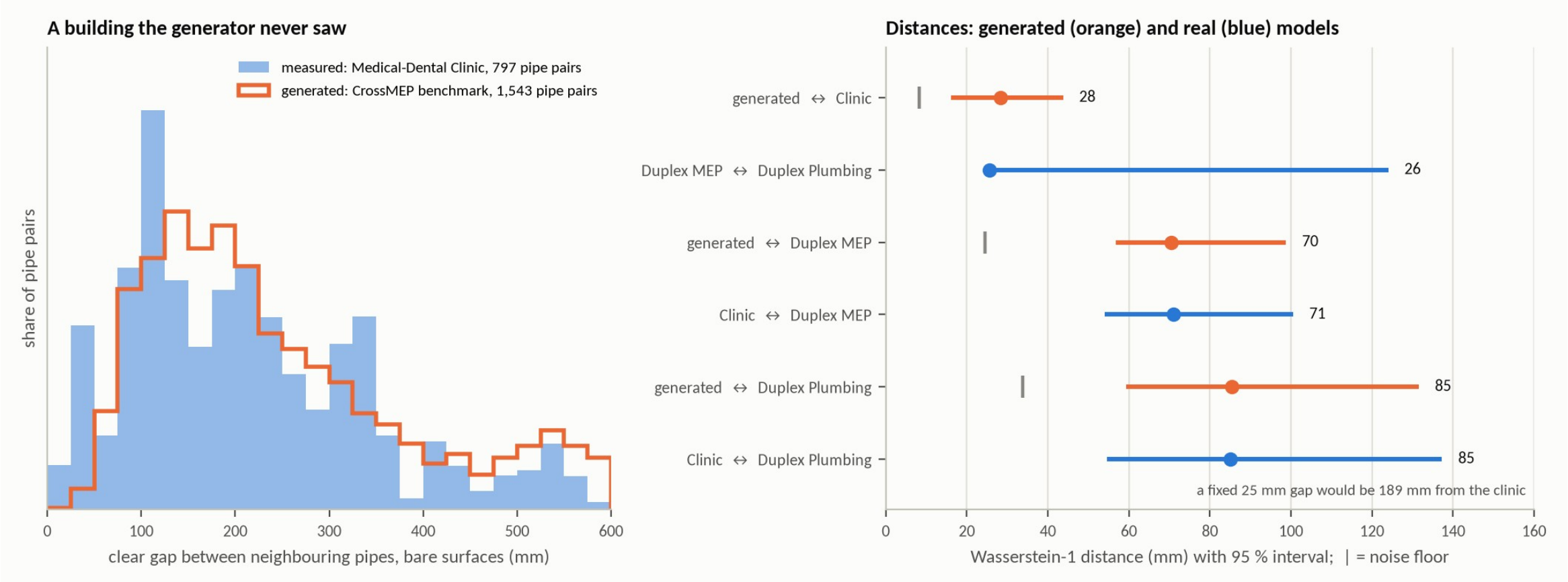
One benchmark context per tier; dashed rings are insulation. Within a tier, kinds, trades, services, surfaces and stacking all vary, so element count is the one controlled difficulty axis. C1 is the most common support in any building; C8 is a congested rack.

Higher tiers are heavier and, overall, tighter, as designed



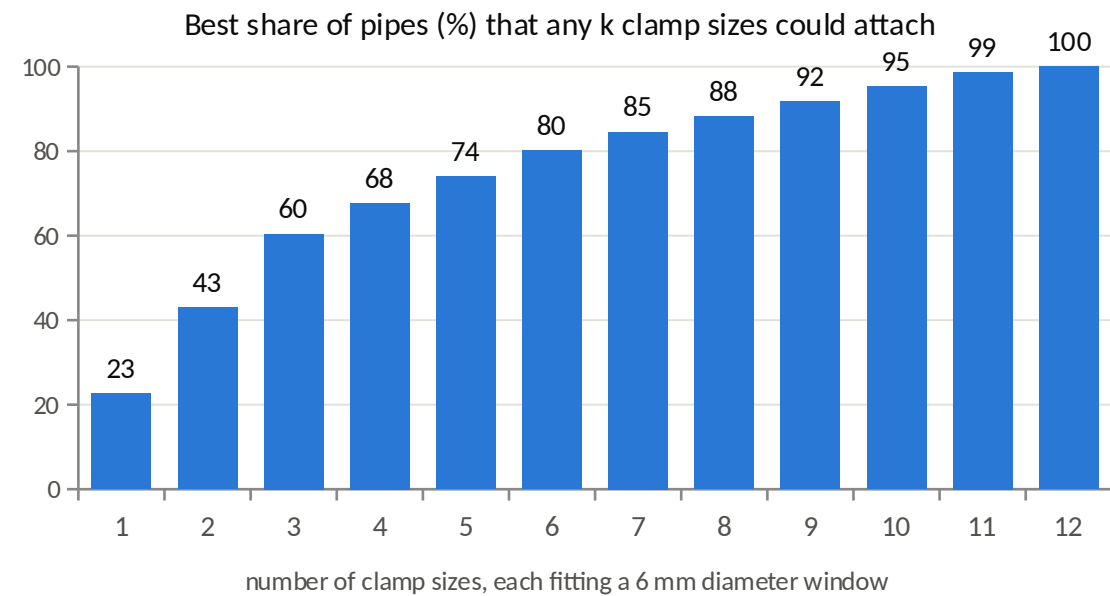
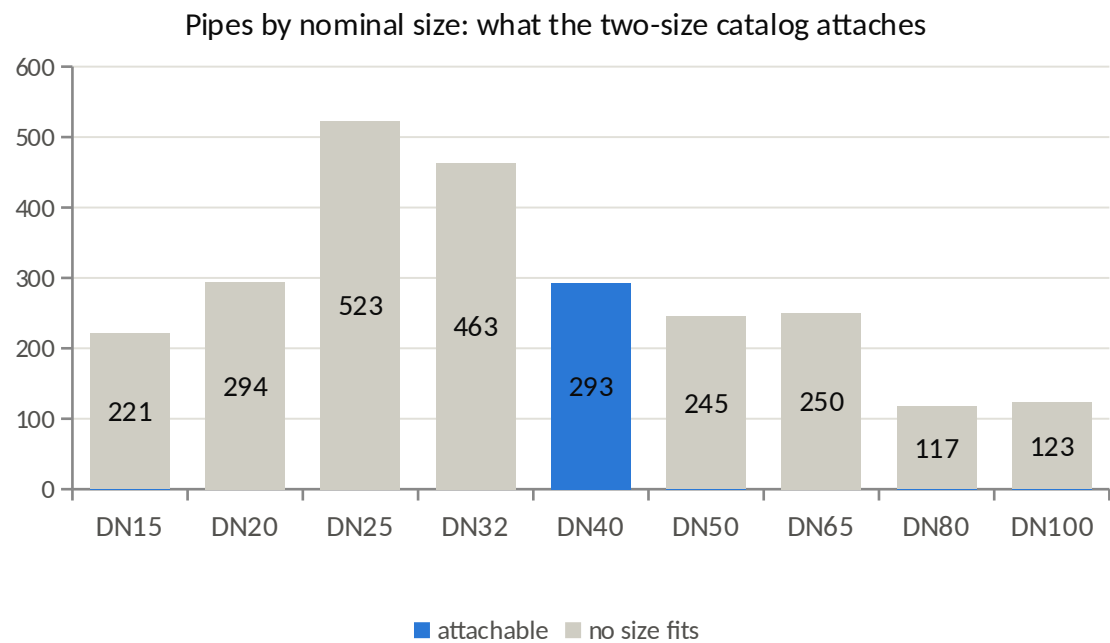
A check that the dataset behaves as designed: the tier fixes the element count; load and congestion follow from the rules. Boxes: benchmark, 125 contexts per tier (IQR, whiskers 1.5 IQR); diamonds: medians of 2,000 generated contexts per tier. Load rises at every tier (rank correlation +0.56); the closest gap shrinks overall (-0.36) but widens at C6 (70 → 75 mm), where elements start to stack in two or three rows.

Spacing holds up on a clinic the generator never saw



The gap distribution was fitted on the duplex; the clinic was held out. Clear gaps between side-by-side pipes, on sections every 250 mm. Clinic (797 pipe pairs): 28 mm from the generated gaps (95 % CI 16–44), more than the 8 mm a perfect generator would show, and about as close as the duplex’s own two models are to each other (26 mm). Models: buildingSMART Duplex Apartment and Medical-Dental Clinic, CC BY 4.0.

A two-size catalog attaches 1 pipe in 9; the dataset shows what to cover



11.6 %

of benchmark pipes attachable with the paper's two sizes (95 % CI 10.1–13.0); every one is DN40

43.0 %

with two sizes placed where the pipes are; 80 % with six, and every pipe with 12

0 of 1,971

trays, ducts and conduits: a pipe-clamp catalog defines no attachment. Load never binds (0.88 vs 2.5 kN)

Set the parameters and the generator returns the designer's brief

SERVICES AT THE HANGER

By tier Exact mix

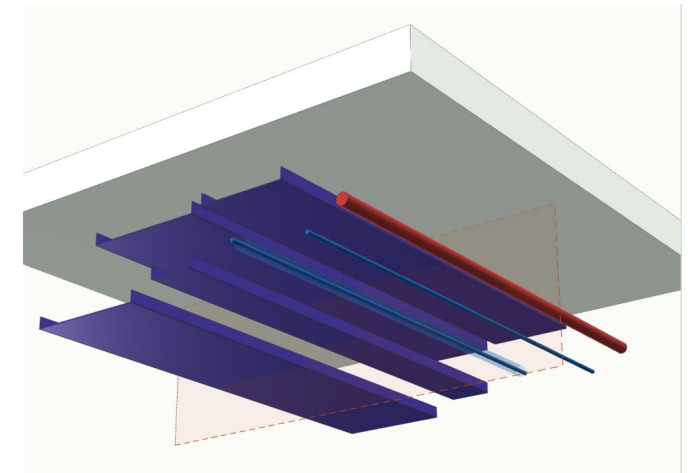
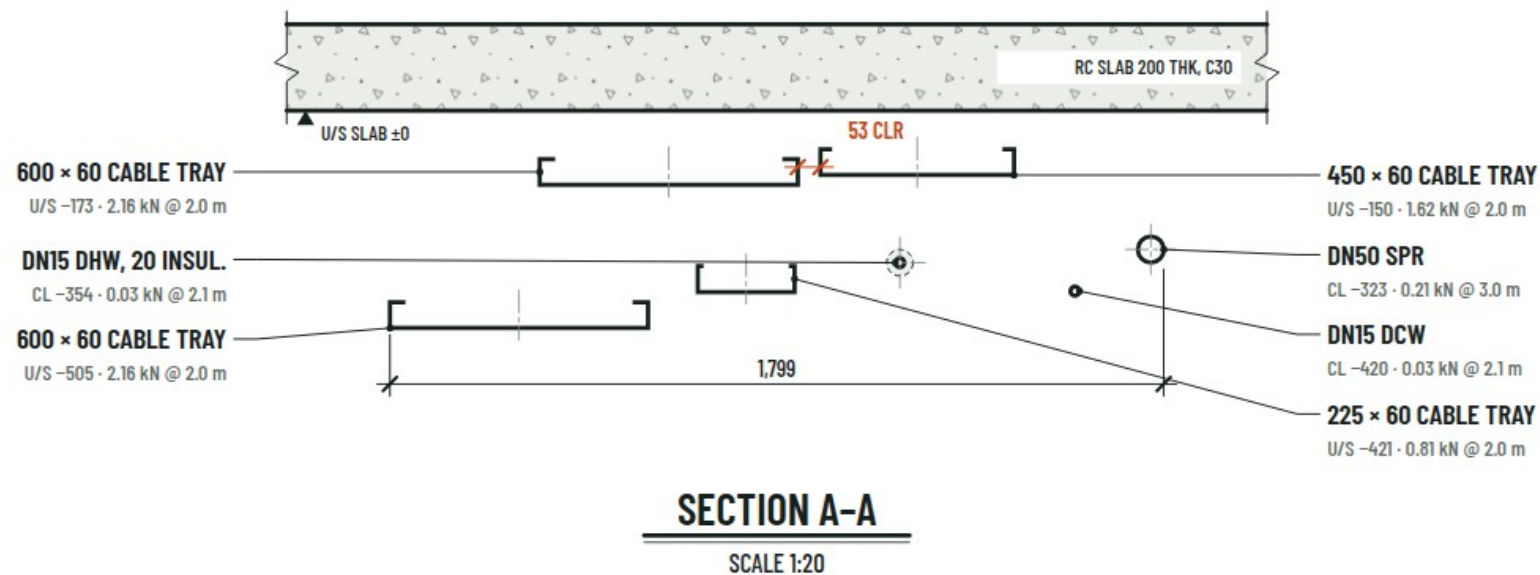
TIER = NUMBER OF SERVICES

C1 C2 C3 C4 C5 C6 C7 C8

SEED

< 0 >

Generate another



Scan to open the studio

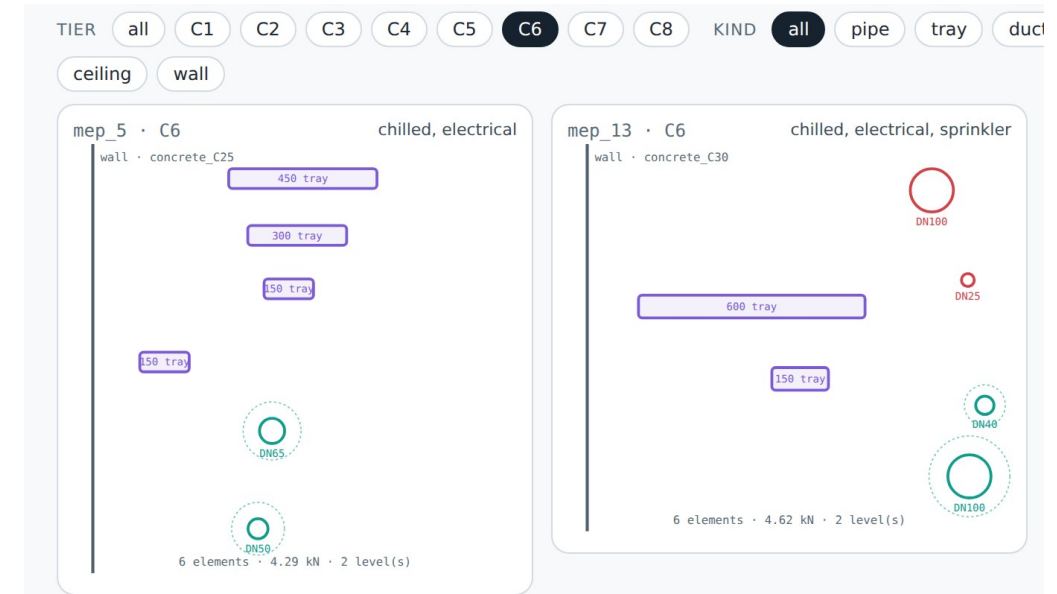
In: the parameters (tier C7, seed 0). Out: section A-A at the support, as the support designer receives it (scale 1:20), and the same run in 3D. Every section is a stored output of the released generator, with the Python call that reproduces it.

Train on the generator, evaluate on the benchmark, report per tier

```
import crossmep.tasks as cm
data = cm.load("benchmark") # 1,000 contexts
cm.filter_contexts(data, pipes=2, trays=1)

from crossmep.evaluate import score, compare
score(my_results, data) # per tier, 95 % CI
compare(mine, baseline, data) # paired test
```

Loading, metrics and scoring need only the Python standard library; the generator needs NumPy. Command line: `python -m crossmep generate | validate | evaluate`.



Interactive gallery: filter the benchmark by tier, kind, trade and surface.

Train

on the generator: unlimited seeds, a curriculum from C1 up to C8

Evaluate

on the benchmark: 125 held-out contexts per tier, the same for every method

Report

per tier with 95 % intervals, paired tests between methods

One section at one support today; a checker and real data come next

Scope

- One cross-section at one support: routing, branches and elevation changes are outside it; so is support spacing — the span is recorded per element, so a method can vary it
- Realism checked on sections of two open buildings, a residential duplex and a medical clinic; congested racks rest on coordination practice and standards
- Trade mix and tier composition are design choices, not survey results
- Synthetic: not for the structural design of real installations

Next

- A public checker for support designs, with best-known costs from an exact solver on the small tiers
- The catalog as an input: generated catalogs, and tests on catalogs a method has never seen
- A real test set: sections from commercial projects with supports designed by engineers

If you coordinate services or design supports, open the gallery, filter for the scenes you know, and tell us what looks wrong. Every rule is one documented constant away from being changed.



Support design now has open problems to learn from

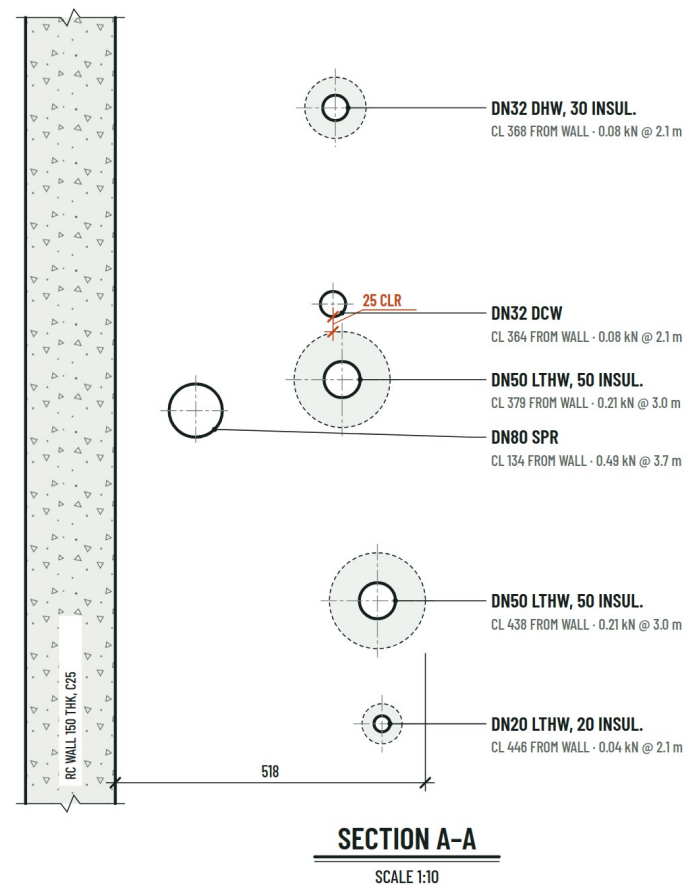
7,000 support-design problems · every constant sourced or declared
pipe gaps checked on two open buildings · open data and code

Data CC BY 4.0, code MIT. Not for the structural design of real installations.

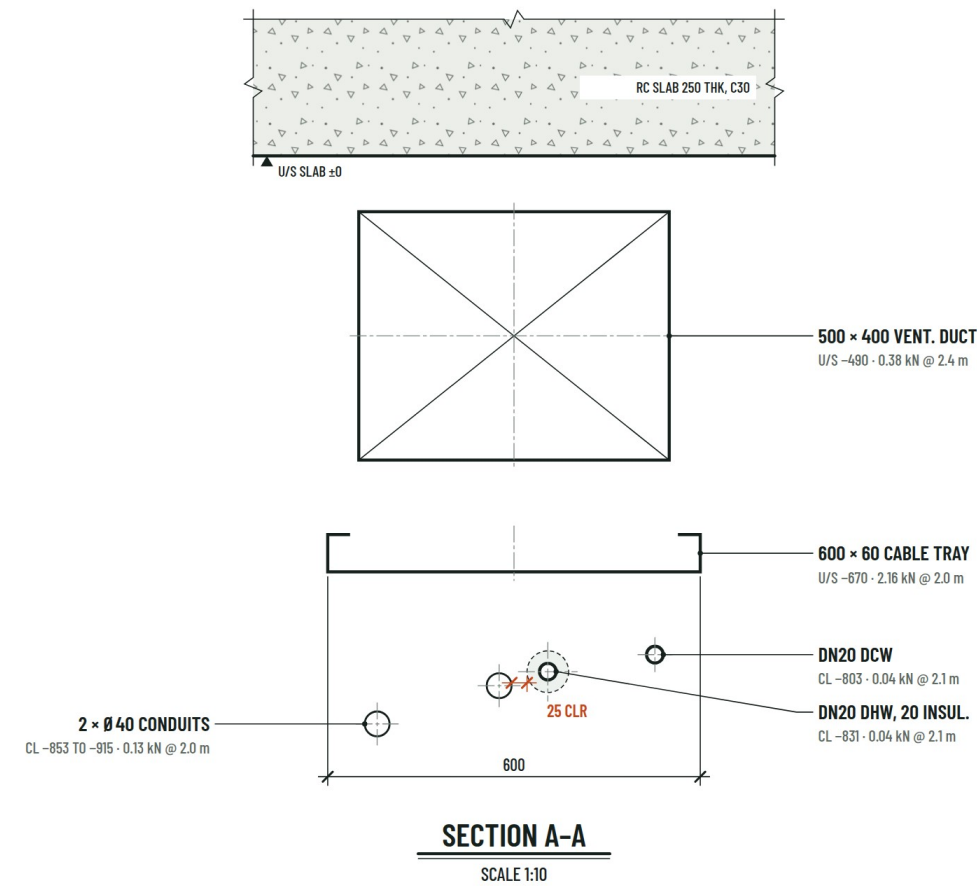


github.com/lavinia-ped/CrossMEP_Dataset
lavinia@stanford.edu

A wall with six pipes, and a slab with a duct, a tray and conduits



Tier C6, seed 0, section 5: six pipes fixed to a wall; offsets from the wall face, the closest clear gap 25 mm.



Exact mix: 2 pipes, 1 tray, 1 duct, 2 conduits on a slab; levels below the soffit and the load at the support.

Median load rises from 0.10 kN at C1 to 1.71 kN at C8 as the gap narrows

tier	contexts	median clear gap (mm)	median load (kN)	median width (mm)
C1	125	—	0.10	102
C2	125	120	0.23	344
C3	125	98	0.44	613
C4	125	90	0.55	728
C5	125	73	0.78	1027
C6	125	76	1.21	993
C7	125	60	1.49	1138
C8	125	62	1.71	1448

4,500 elements

2,529 pipes · 1,141 conduits · 569 trays · 261 ducts

28 mm

Wasserstein-1, generated to measured pipe gaps of the clinic
(95 % CI 16–44)

–0.36 / +0.56

rank correlation of tier with the closest gap / with the load

Against the clinic the generator is 28 mm off; the small duplex is 70 to 85

pipe gaps	pairs	sections	median (mm)	< 25 mm
generated (benchmark)	1,543	—	207	0 %
Clinic, Plumbing	797	3,833	203	2 %
Duplex, MEP	74	204	177	23 %
Duplex, Plumbing	41	112	114	32 %

- Sections every 250 mm; runs within 400 mm of height form a row; gaps between bare pipe surfaces, below 600 mm
- Robust: across 11 settings of the cut (spacing, row band, axis tolerance, run length) generated ↔ clinic stays at 27–36 mm
- Noise floor: the distance a perfect generator would still show at that sample size
- The duplex samples are small, hence the wide intervals

Wasserstein-1 (mm)	by length	each pair once	noise floor
generated ↔ Clinic	28 (16–44)	15 (11–24)	8
generated ↔ Duplex MEP	70 (57–99)	94 (74–119)	25
generated ↔ Duplex Plumbing	85 (59–132)	100 (85–125)	34
Clinic ↔ Duplex MEP	71 (54–101)	88 (70–118)	—
Clinic ↔ Duplex Plumbing	85 (55–137)	94 (77–121)	—
Duplex MEP ↔ Duplex Plumbing	26 (24–124)	25 (21–108)	—

Intervals: 95 % bootstrap over contexts (generated) and pipe pairs (measured). Models: buildingSMART Duplex Apartment and Medical-Dental Clinic, CC BY 4.0; derived measurements and attribution in [verify/measured/](#).

The diameter rule moves catalog coverage more than tier or split does

tier	pipes	benchmark % (95 % CI)	population % (95 % CI)
C1	71	11.3 (4.1–19.1)	11.4 (9.6–13.3)
C2	147	12.2 (6.5–18.8)	10.9 (9.4–12.5)
C3	203	10.3 (5.7–15.6)	10.8 (9.6–12.0)
C4	257	13.6 (8.4–19.3)	10.4 (9.4–11.4)
C5	391	9.2 (6.1–12.6)	10.5 (9.6–11.5)
C6	410	11.5 (7.9–15.4)	10.8 (9.9–11.8)
C7	465	10.8 (7.5–14.1)	11.0 (10.1–11.8)
C8	585	13.3 (9.8–17.1)	11.2 (10.4–12.0)

attach diameter rule	bins ± 0 mm	bins ± 0.5 mm	bins ± 1 mm
service (paper)	11.6	12.9	12.9
bare	11.6	16.4	16.4
insulated	6.2	7.5	7.5

% of benchmark pipes attachable when both bin edges are widened by the stated amount. DN100 (114.3 mm) lies 0.3 mm above the 108–114 mm bin, so the headline moves with the diameter rule and the edge tolerance.

split	pipes	% attachable (95 % CI)
train	12,930	10.8 (10.2–11.5)
val	1,254	9.4 (7.4–11.4)
test	1,271	9.0 (7.2–11.0)
benchmark	2,529	11.6 (10.1–13.0)

- Intervals: cluster bootstrap over contexts (2,000 resamples), because the elements of one context are correlated
- Population: 2,000 freshly generated contexts per tier; overall 10.8 % (95 % CI 10.5–11.2)
- Only size and capacity are tested, so coverage is an upper bound on what can be installed
- Best possible share with k sizes: exact optimum over the distinct attach diameters

A context is stored as plain JSON: elements, sizes, loads, positions

```
{
  "context_id": "mep_57", "tier": "C2",
  "surface": {
    "kind": "ceiling", "substrate": "concrete_C25", "thickness_mm": 200.0
  },
  "n_elements": 2, "n_levels": 1, "total_load_kN": 1.12, "bundle_width_mm": 588.0,
  "elements": [
    {
      "kind": "cable_tray", "service": "tray", "trade": "electrical", "shape": "rect",
      "label": "300 tray", "width_mm": 300.0, "height_mm": 60.0, "insulation_mm": 0.0,
      "load_kN": 1.08, "span_m": 2.0, "load_kN_per_m": 0.5396,
      "level": 0, "along_mm": -144.0, "out_mm": 120.0
    },
    {
      "kind": "pipe", "service": "heating", "trade": "heating", "shape": "round",
      "label": "DN20", "width_mm": 26.9, "height_mm": 26.9, "insulation_mm": 20.0,
      "load_kN": 0.04, "span_m": 2.1, "load_kN_per_m": 0.0189,
      "level": 0, "along_mm": 260.5, "out_mm": 156.2
    }
  ]
}
```

- Units: millimetres, kilonewtons, metres (span)
- Widths are bare; insulation_mm is per side
- $\text{load_kN} = \text{load_kN_per_m} \times \text{span_m}$
- along_mm, out_mm: position along and out from the surface
- level: the generative row (metadata)